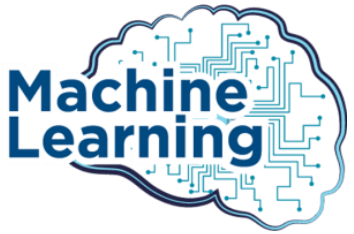


Chapter-3
Unsupervised Learning

Overview of Unsupervised Learning



- Unsupervised learning is a type of machine learning that involves training a model on data without explicit labels.
- Instead, the algorithm identifies/discovers hidden patterns, structures, or relationships in the data.
- Common unsupervised learning approaches:
  - clustering,
  - association rule mining,
  - dimensionality reduction.

Unsupervised Learning Application

Machine Learning

01 **Computer vision:**

object recognition.

02 Market Basket Analysis

Finding associations between items in transactional data.

03 **Anomaly detection**

Discover atypical data points within a dataset(Identifying data points that deviate significantly from the norm)

04 **Customer personas**

Defining customer personas makes it easier to understand common traits and business clients' purchasing habits. Used for Advertisement, ad on

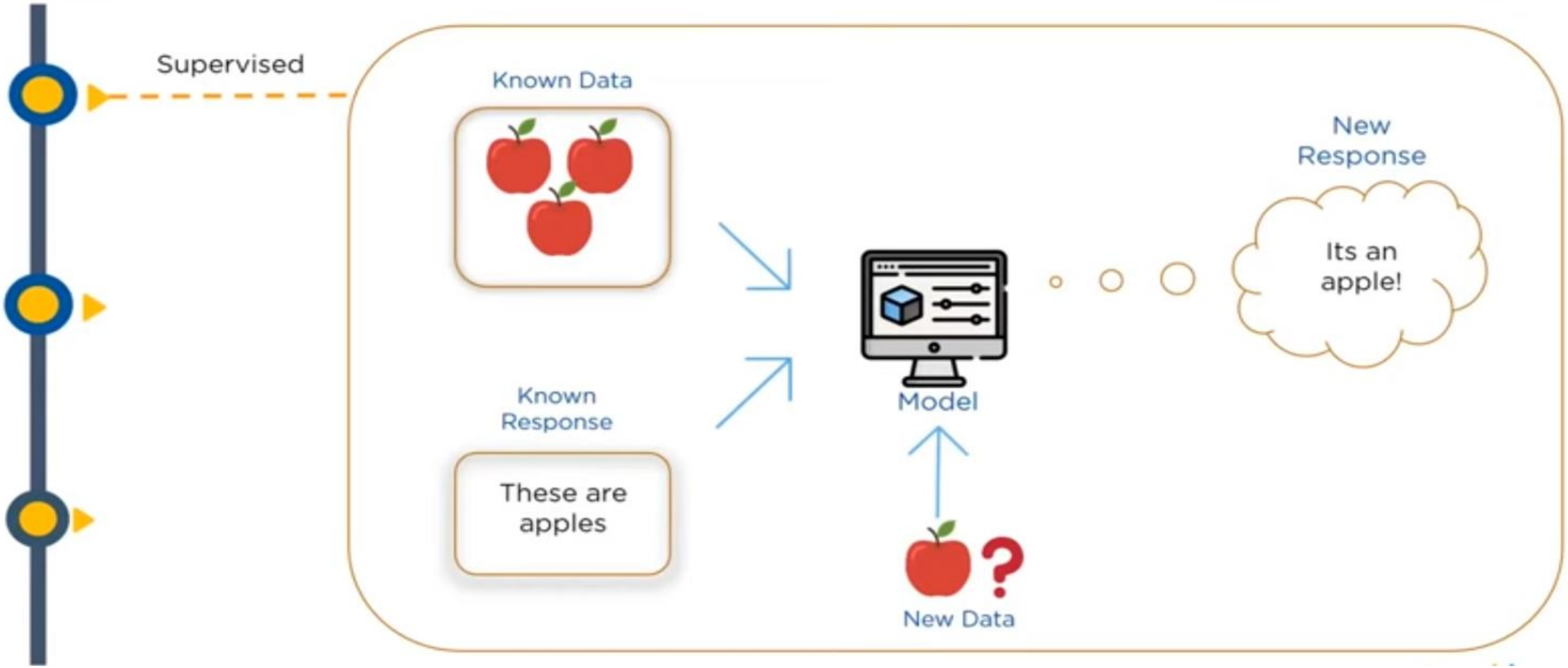
05 **Recommendation engine**

Using past purchase behavior data, develop more effective cross-selling strategies.

06 Dimensionality Reduction

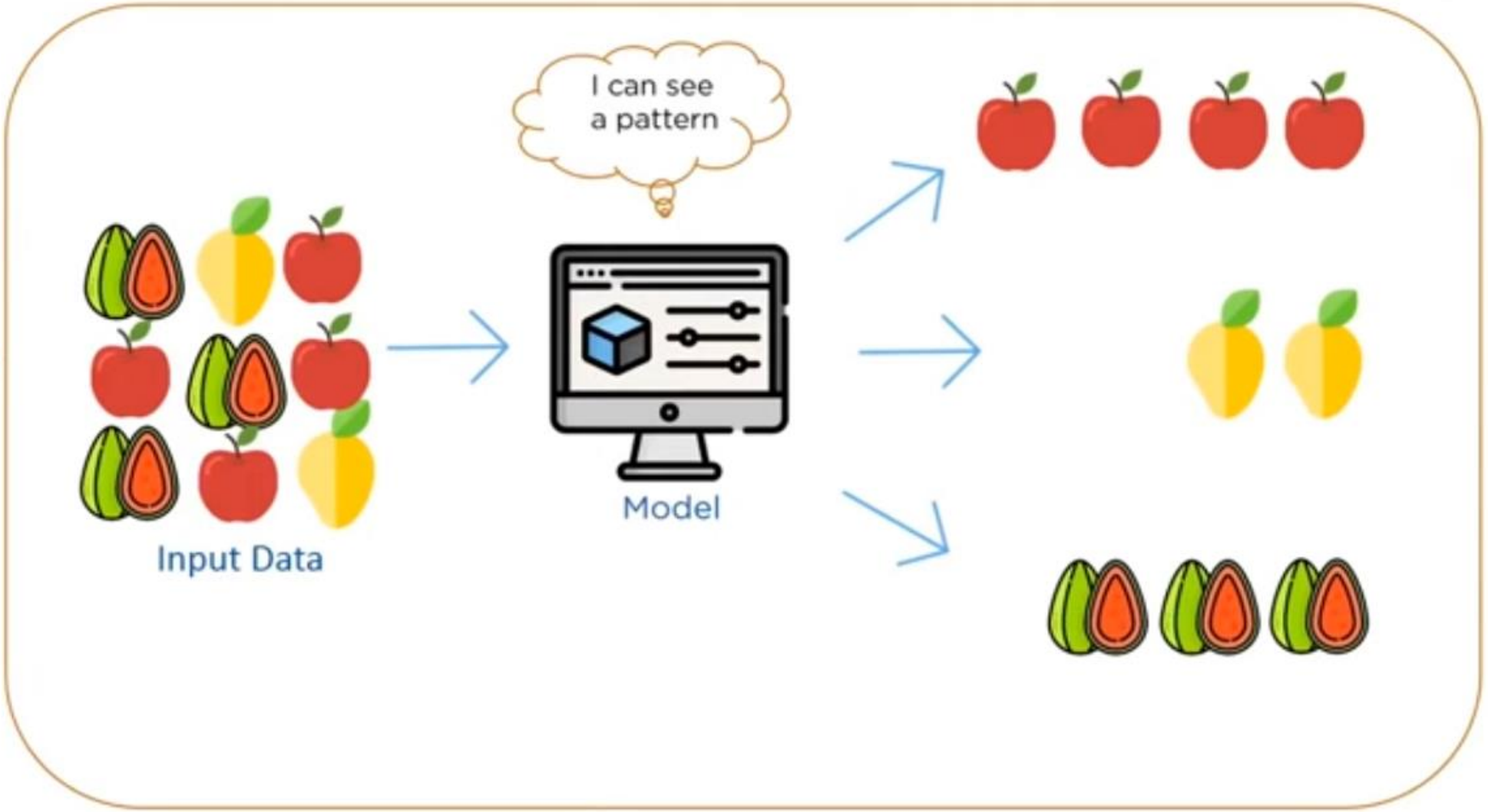
Reducing the number of features in a dataset while preserving its structure.

Recap

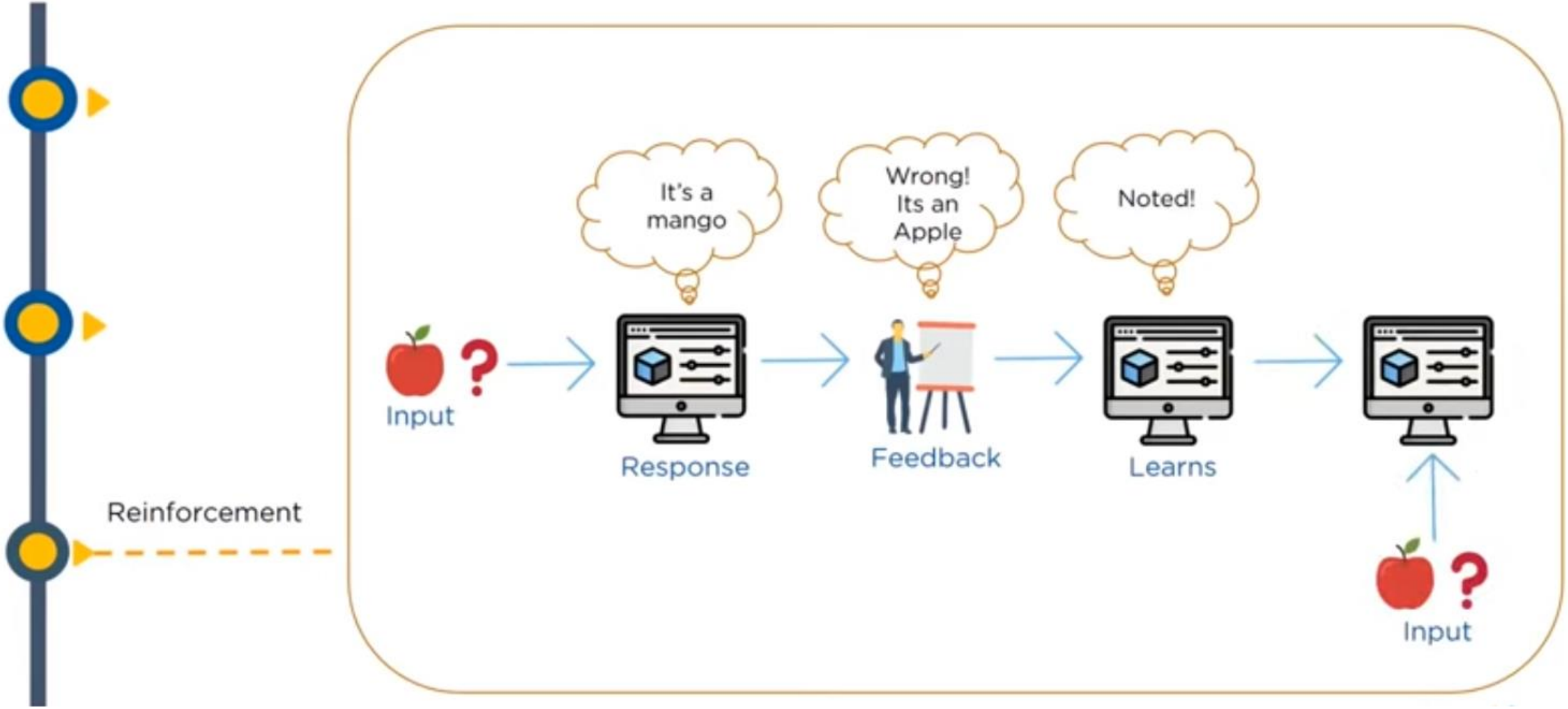

Supervised
Known Data
Known Response
These are apples
Model
New Data
New Response
Its an apple!

Recap

Unsupervised



Recap


Reinforcement
Input
It's a mango
Response
Wrong! Its an Apple
Feedback
Noted!
Learns
Input

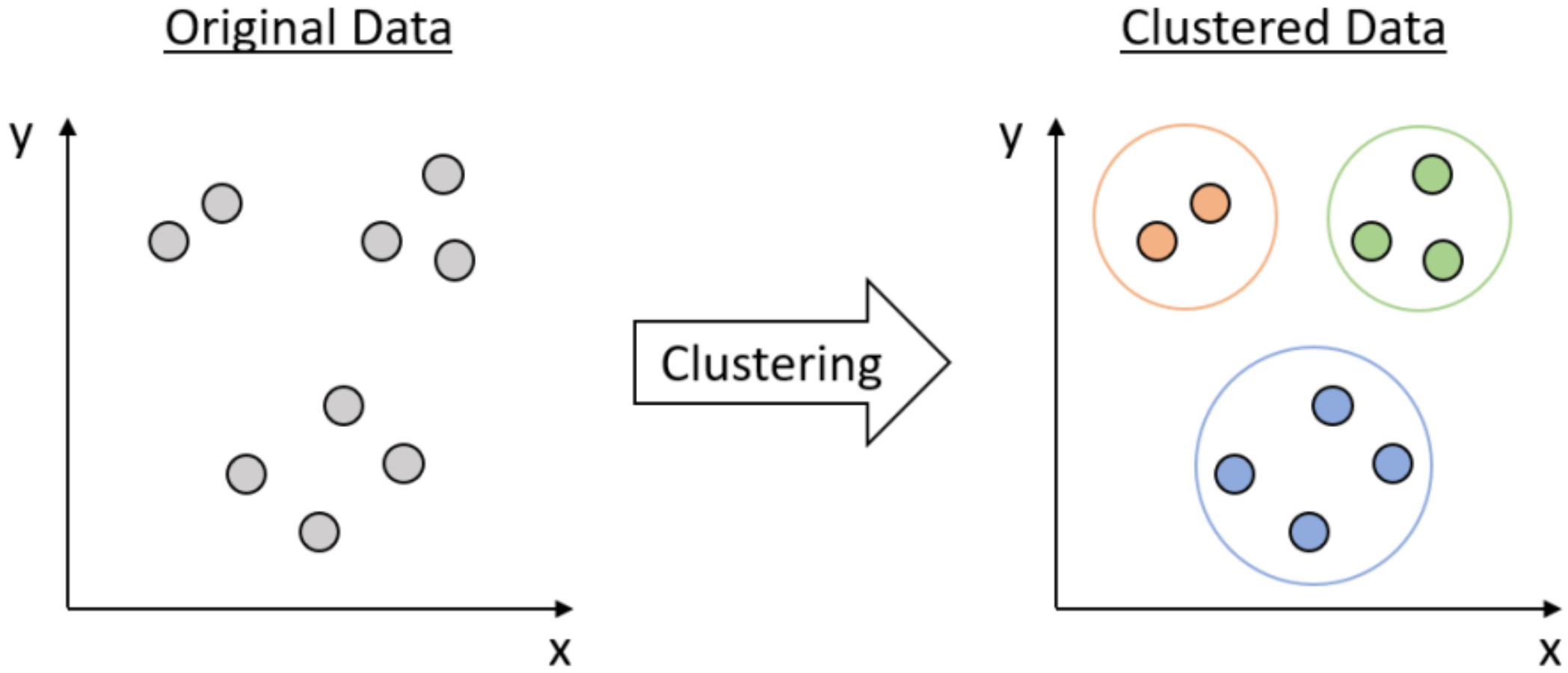
What is Clustering?

❖Clustering is the process of dividing the datasets into groups, consisting of similar data points.

- Data Points in the same group are as similar as possible.
- Data Points in different group are as different as possible.

❖Goal of clustering is to determine the intrinsic grouping in a set of unlabeled data.

What is Clustering?



Cont'd

- Objective:
  - Identify natural groupings in the data.
  - Minimize intra-cluster distance (within clusters).
  - Maximize inter-cluster distance (between clusters).
- Input:
  - Unlabeled dataset (e.g., feature vectors).
- Output:
  - A partitioning of the data into clusters or a hierarchical structure of clusters.

Where clustering used?



Retail Store

analyzing customer shopping behaviour, and customer retention.



Banking

Fraud Detection,
Customer segmentation
Credit Scoring
customer retention.



Insurance Companies

Fraud Detection,
customer retention.

amazon

Recommendation

NETFLIX

Recommended Movies

flickr

Flickr's Photos

Types of Clustering

❖There are many different clustering algorithms as there are multiple ways to define a cluster. Different approaches will work well for different types of models depending on the size of the input data, the dimensionality of the data, the rigidity of the categories and the number of clusters within the dataset.

- Centroid-based(partitioning clustering)
- Hierarchical clustering
- Density based clustering

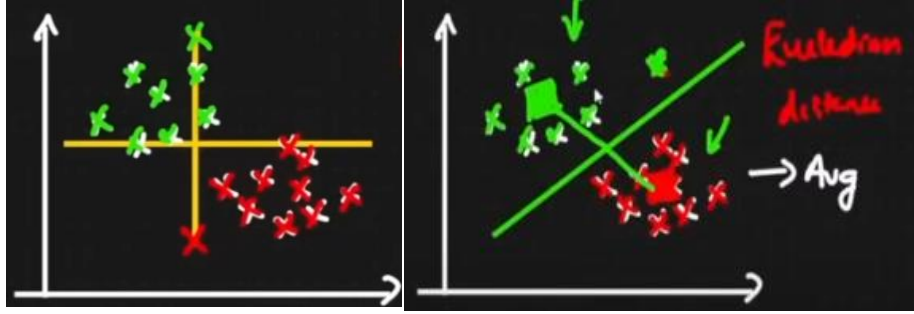
1. Centroid-based clustering

❖Centroid-based clustering is a type of clustering method that partitions or splits a data set into similar groups based on the distance between their centroids.

❖ Each cluster’s centroid, or center, is either the mean or median of all the points in the cluster depending on the data.

❖Best example of centroid based clustering is k-means clustering.

K-Means Clustering Algorithm



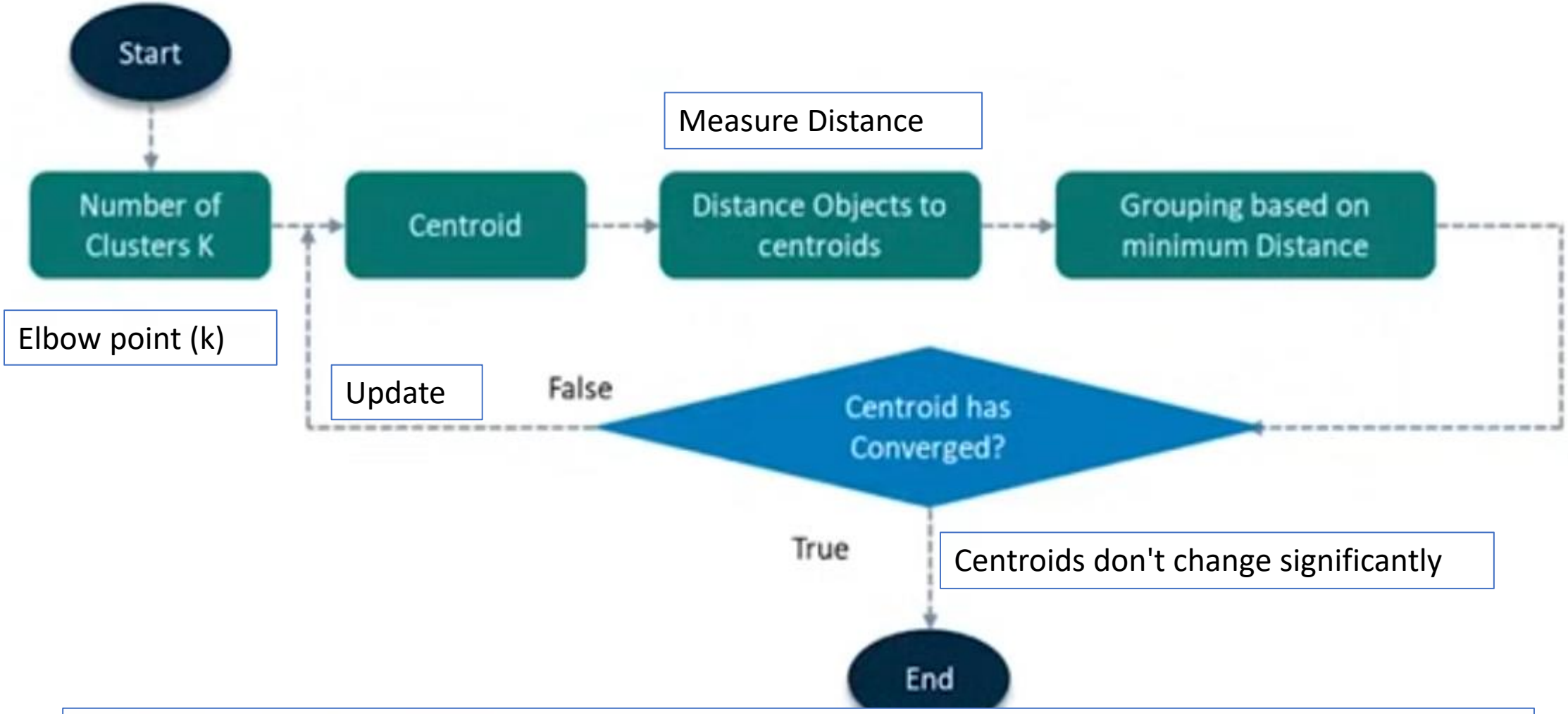
•Divides data into k clusters.

Steps:

1. Decide the number of clusters to be made(Guessing)
2. Initialize k cluster centroids randomly(Pick up k random points as cluster center called centroid.).
3. Calculate the Euclidean distance between the data points and each centroids. Assign each data point to the nearest centroid.
4. Update(find new) centroids based on the average of points in each cluster.
5. Repeat step 2, 3, and 4 until convergence (cluster don't change significantly).

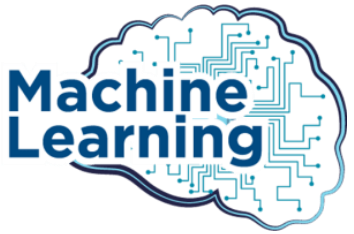
The Euclidean distance is used to find out which centroid is closest to each data point and the data points are assigned to the corresponding centroids.

How Clustering Algorithm works



Distance measure will determine the similarity between two elements

K-means algorithm - solved example



- Apply K Means clustering algorithm to cluster the following data points into two groups.
- Data Points: -(2, 10), (2, 5), (8, 4), (5, 8), (7, 5), (6, 4) , (1, 1), (4, 9)
- The distance function used is Euclidean distance.
- Use, three data points((2, 10), (5, 8), (1, 2)) as initial centroids.

K-means algorithm - solved example

Initial Centroids:
A1: (2, 10)
B1: (5, 8)
C1: (1, 2)

Data Points			Distance to						Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10	0.00		3.61		8.06		1	
A2	2	5	5.00		4.24		3.16		3	
A3	8	4	8.49		5.00		7.28		2	
B1	5	8	3.61		0.00		7.21		2	
B2	7	5	7.07		3.61		6.71		2	
B3	6	4	7.21		4.12		5.39		2	
C1	1	2	8.06		7.21		0.00		3	
C2	4	9	2.24		1.41		7.62		2	

New Centroid

A1: (2, 10)

B1: (6, 6)

C1: (1.5, 3.5)

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

New Centroid=average of the distance in a given cluster e.g. C1=((2+1)/2, (5+2)/2)=(1.5, 3.5)

K-means algorithm - solved example

Current Centroids:
A1: (2, 10)
B1: (6, 6)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			2	10	6	6	1.5	1.5		
A1	2	10	0.00		5.66		6.52		1	1
A2	2	5	5.00		4.12		1.58		3	3
A3	8	4	8.49		2.83		6.52		2	2
B1	5	8	3.61		2.24		5.70		2	2
B2	7	5	7.07		1.41		5.70		2	2
B3	6	4	7.21		2.00		4.53		2	2
C1	1	2	8.06		6.40		1.58		3	3.
C2	4	9	2.24		3.61		6.04		2	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-means algorithm - solved example

Current Centroids:
A1: (2, 10)
B1: (6, 6)
C1: (1.5, 3.5)

New Centroids:
A1: (3, 9.5)
B1: (6.5, 5.25)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			2	10	6	6	1.5	1.5		
A1	2	10	0.00		5.66		6.52		1	1
A2	2	5	5.00		4.12		1.58		3	3
A3	8	4	8.49		2.83		6.52		2	2
B1	5	8	3.61		2.24		5.70		2	2
B2	7	5	7.07		1.41		5.70		2	2
B3	6	4	7.21		2.00		4.53		2	2
C1	1	2	8.06		6.40		1.58		3	3
C2	4	9	2.24		3.61		6.04		2	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-means algorithm - solved example

Current Centroids:
A1: (3, 9.5)
B1: (6.5, 5.25)
C1: (1.5, 3.5)

New Centroids:
A1: (3.67, 9)
B1: (7, 4.33)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			3	9.5	6.5	5.25	1.5	3.5		
A1	2	10	1.12		6.54		6.52		1	1
A2	2	5	4.61		4.51		1.58		3	3
A3	8	4	7.43		1.95		6.52		2	2
B1	5	8	2.50		3.13		5.70		2	1
B2	7	5	6.02		0.56		5.70		2	2
B3	6	4	6.26		1.35		4.53		2	2
C1	1	2	7.76		6.39		1.58		3	3
C2	4	9	1.12		4.51		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-means algorithm - solved example

Current Centroids:
A1: (3.67, 9)
B1: (7, 4.33)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			3.67	9	7	4.33	1.5	3.5		
A1	2	10	1.94		7.56		6.52		1	1
A2	2	5	4.33		5.04		1.58		3	3
A3	8	4	6.62		1.05		6.52		2	2
B1	5	8	1.67		4.18		5.70		1	1
B2	7	5	5.21		0.67		5.70		2	2
B3	6	4	5.52		1.05		4.53		2	2
C1	1	2	7.49		6.44		1.58		3	3
C2	4	9	0.33		5.55		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Now, all the clusters are converged

Evaluation of K-Means Clustering

- Validating the quality of K-Means clustering is important to ensure that the clusters properly represent the data structure.
- Since clustering is unsupervised, you typically don't have ground truth labels to compare against.
  - Evaluating the quality of clusters can be challenging.

Evaluation of k-means clustering - Internal Metrics

❖ These metrics evaluate the quality of clustering based on the data itself, without requiring ground truth labels.

a) Silhouette Score: Measures how similar data points are to their own cluster compared to other clusters.

❖ Range: -1 to 1.

- 1: Perfectly clustered (points are far from other clusters).
- 0: Points are on or near the decision boundary between clusters.
- -1: Points are in the wrong cluster.

❖ For each point:

- a = average distance to other points in the same cluster.
- b = average distance to points in the nearest neighboring cluster.
- Silhouette score = $(b-a)/\max(a, b)$

❖ Goal: Higher silhouette score indicates better clustering.

b). Inertia (Within-Cluster Sum of Squares(WSS)): Measures compactness of clusters (used in K-Means).

How to Determine the Optimal Number of Clusters (k)?

❖Choosing the right number of clusters is a critical part of K-Means clustering.

Below are some techniques to find the optimal value of k:

a) Elbow Method

- Plot the inertia (WCSS) against the number of clusters (k).
- Look for the "elbow point," where the rate of decrease in inertia slows down.
- This indicates the optimal number of clusters.

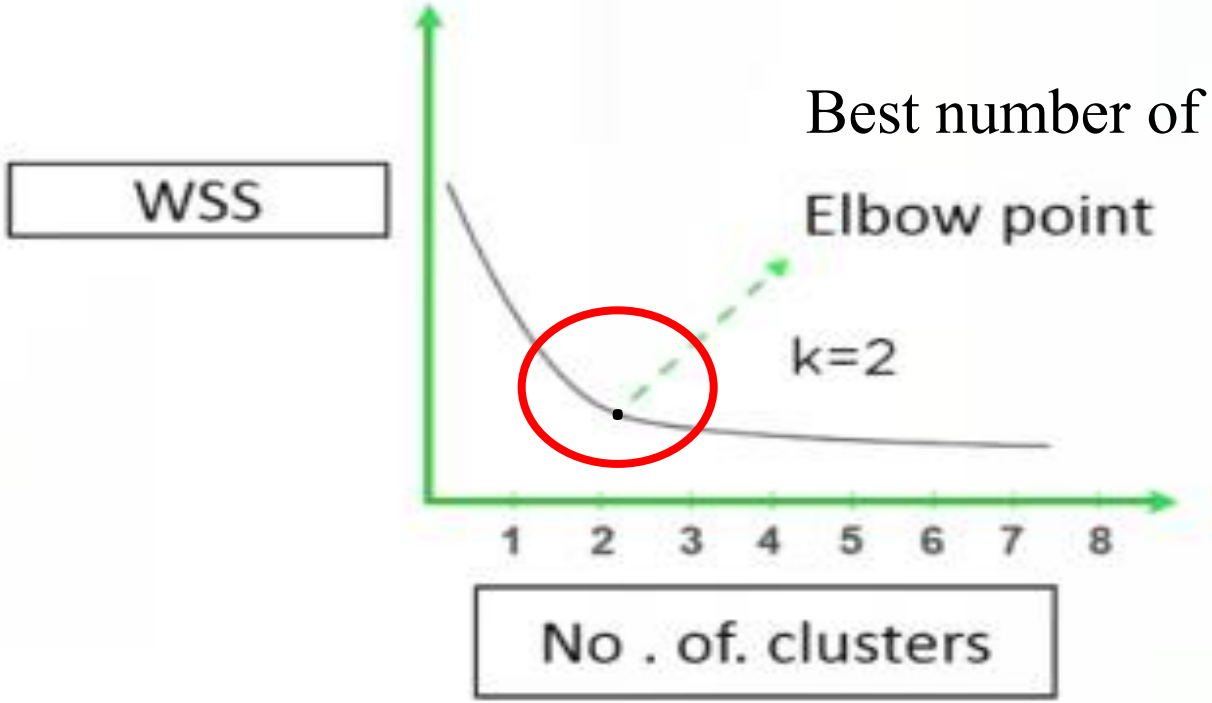
Elbow Method

Within sum of squares (WSS) is defined as the sum of the squared distance between each member of the cluster and its centroid

$$WSS = \sum_{i=1}^{m} (x_i - c_i)^2 \qquad = \sum_{i=1}^{K} \sum_{x \in c_i} dist(x, c_i)^2$$

Where x_i = data point and c_i = closest point to centroid

Elbow Method



Best number of clusters at the "Elbow point"

- Now, we draw a curve between *WSS* (within sum of squares) and the *number of clusters*
- Here, we can see a very slow change in the value of WSS after k=2, so you should take that elbow point value as the final number of clusters

Quiz

What is the role of the "K" parameter in K-means clustering?

- It represents the maximum number of iterations allowed for the algorithm.
- It determines the number of clusters that the algorithm will partition the data into.
- It controls the rate of convergence for the algorithm.
- It sets the threshold for the minimum similarity required between data points.

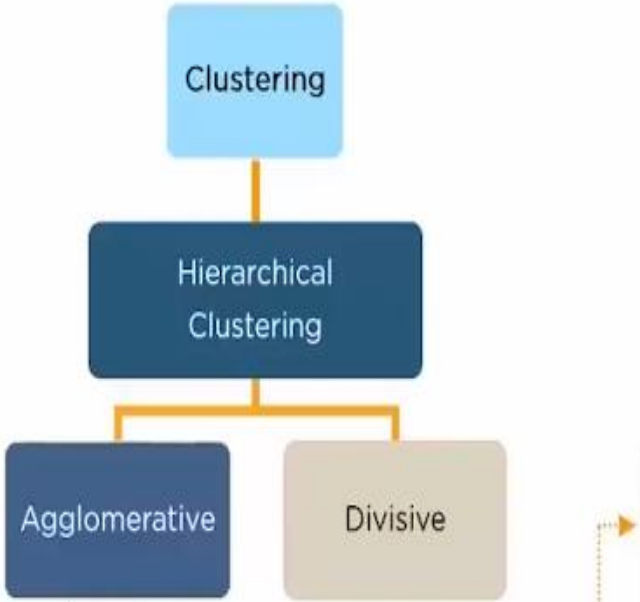
2. Hierarchical clustering

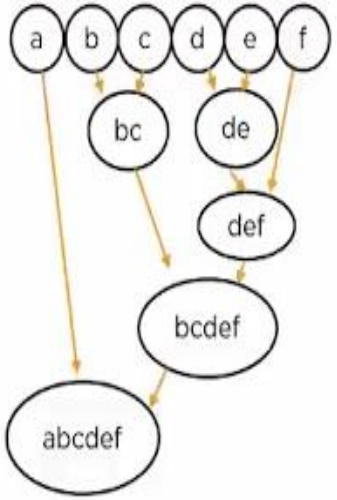
- Hierarchical clustering, sometimes called connectivity-based clustering, groups data points together based on the proximity and connectivity of their attributes.
- This method determines clusters based on how close data points are to one another across all of the dimensions.
- The idea is that objects that are nearer are more closely related than those that are far from each other.

2. Hierarchical clustering(cont'd)

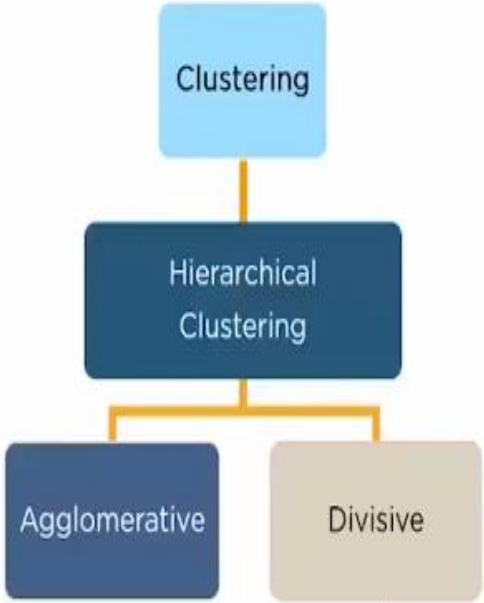
- Unlike k-means, there is no need to pre-specify the number of clusters. Instead, the clustering algorithm creates a graph network of the clusters at each hierarchical level.
- Hierarchical clusters can be graphed with a dendrogram to help visually summarize and organize discovered clusters and the hierarchy that they may contain.
- There are two approaches to performing hierarchical cluster analysis:
  - Agglomerative clustering
  - Divisive clustering

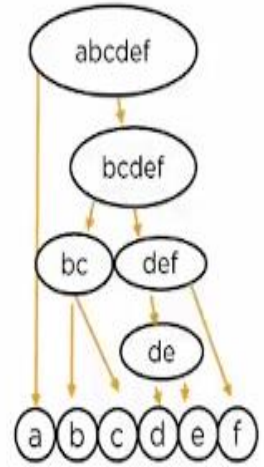
Types of Clustering - Hierarchical





"Bottom up" approach: Begin with each element as a separate cluster and merge them into successively larger clusters





"Top down" approach begin with the whole set and proceed to divide it into successively smaller clusters.

Hierarchical clustering-Agglomerative clustering

- In agglomerative clustering a bottom-up approach starts with individual data points and successively merges clusters by compute the proximity matrix of all the clusters at the current level of the hierarchy to create a tree-like structure.
- There are a variety of choices possible in terms of how these clusters should be merged with one another with tradeoffs in terms of the quality and efficiency of the clustering. In single-linkage clustering, the shortest distance between any pair of data points in two clusters is used as a similarity measure.

Association rule Mining

Given a set of transactions, find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction.

Association rule mining is a technique that shows how items are associated to each other.

Market basket analysis explains the combinations of products that frequently co-occur in transactions.

Example:

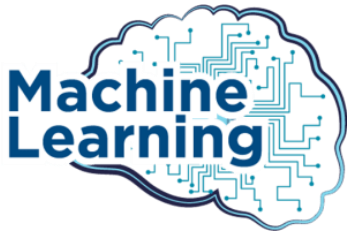
Customer who purchase bread have a 60% likelihood of also purchasing jam.





Customer who purchase laptops are more likely to purchase laptop bags.

Association rule Mining



- These methods are frequently used for market basket analysis, allowing companies to better understand relationships between different products.
- Understanding consumption habits of customers enables businesses to develop better cross-selling strategies and recommendation engines.
- Examples of this can be seen in Amazon's "Customers Who Bought This Item Also Bought" or Spotify's "Discover Weekly" playlist.
- While there are a few different algorithms used to generate association rules, such as Apriori, Eclat, and FP-Growth, the Apriori algorithm is most widely used.

Association rule Mining

Example of Association rule

$$A \Rightarrow B$$

- It means that if a person buys item A then he will also buy item B
- Three common ways to measure association:

- Support
- Confidence
- Lift



Association rule Mining

Support gives fraction of transactions which contains the item A and B

$$Support = \frac{freq(A,B)}{N}$$

Confidence gives how often the items A & B occur together, given no. of times A occurs

$$Confidence = \frac{freq(A,B)}{freq(A)}$$

Lift indicates the strength of a rule over the random co-occurrence of A and B

$$Lift = \frac{Support}{Supp(A)\ x\ Supp(B)}$$

Association Rule Mining Example

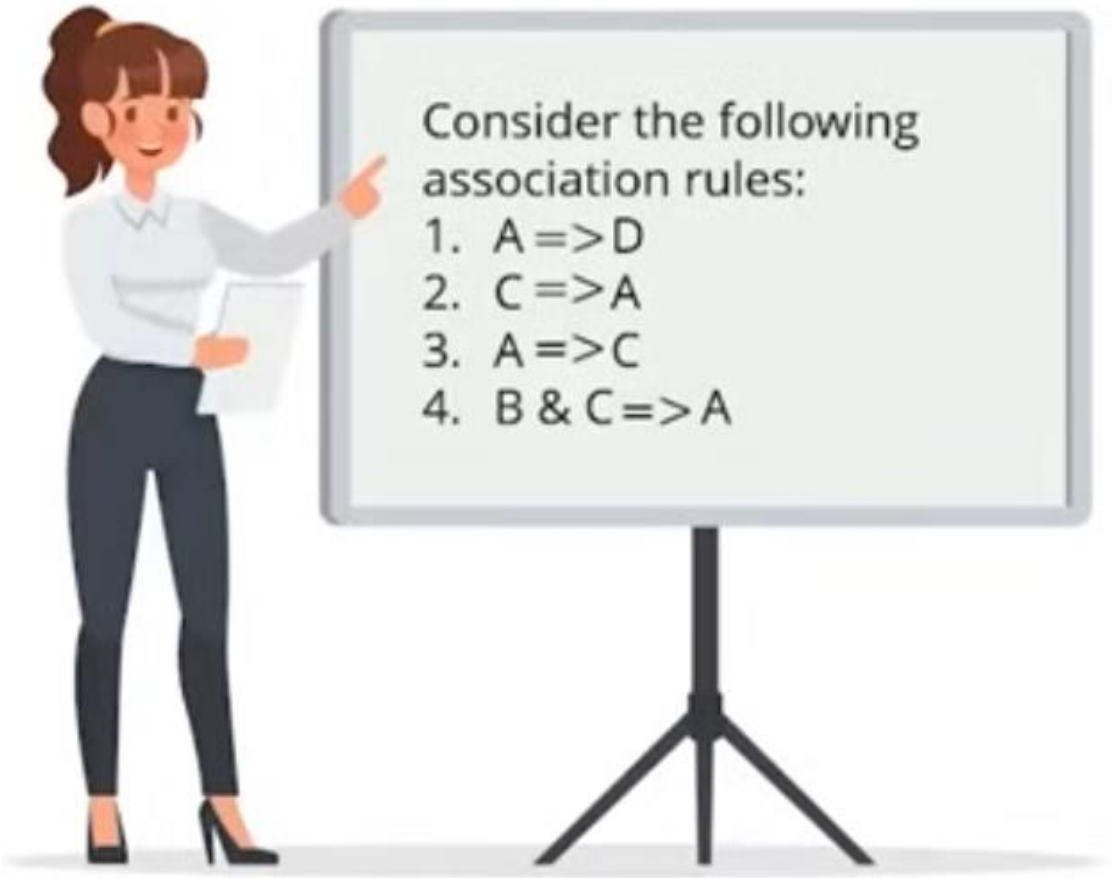
Set of items {A, B, C, D, E}
Set of transactions {T1, T2, T3, T4, T5}

Transactions at a local store

T1	A	B	C
T2	A	C	D
T3	B	C	D
T4	A	D	E
T5	B	C	E



Association Rule Mining Example



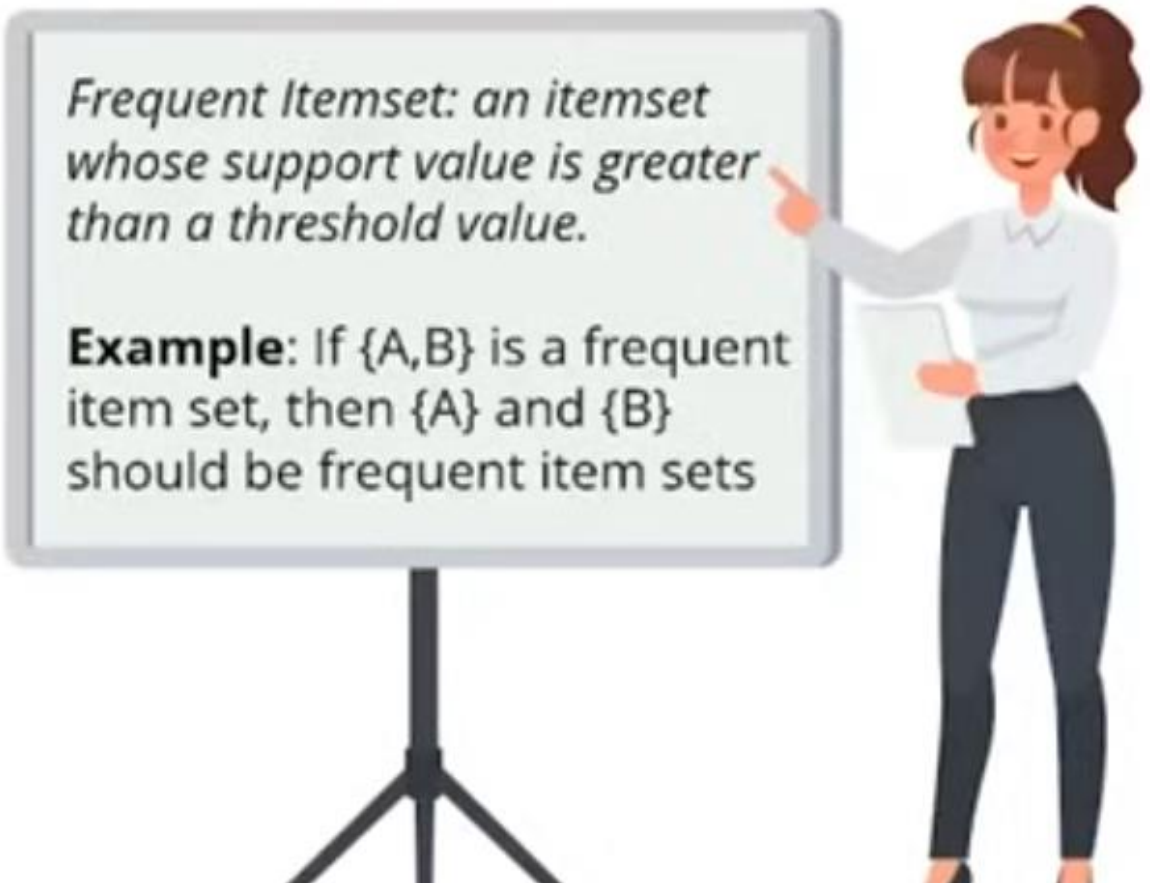
Calculate support, confidence and lift for these rules:

Rule	Support	Confidence	Lift
A=>D	2/5	2/3	10/9
C=>A	2/5	2/4	5/6
A=>C	2/5	2/3	5/6
B, C=>A	1/5	1/3	5/9

Apriori Algorithm

Apriori algorithm uses frequent item sets to generate association rules. It is based on the concept that a subset of a frequent itemset must also be a frequent itemset.





Apriori Algorithm

Consider the following transactions:

TID	Items
T1	1 3 4
T2	2 3 5
T3	1 2 3 5
T4	2 5
T5	1 3 5

Note

Min. support count = 2



Apriori Algorithm First Iteration

Step 1: Create item sets of size one & calculate their support values

TID	Items
T1	1 3 4
T2	2 3 5
T3	1 2 3 5
T4	2 5
T5	1 3 5



Table: C1 |

Itemset	Support
{1}	3
{2}	3
{3}	4
{4}	1
{5}	4



Table: F1 |

Itemset	Support
{1}	3
{2}	3
{3}	4
{5}	4

Item sets with support value less than min. support value (i.e. 2) are eliminated

Apriori Algorithm second Iteration

Step 2: Create item sets of size two & calculate their support values.
All the combinations of item sets in F1 | are used in this iteration

TID	Items
T1	1 3 4
T2	2 3 5
T3	1 2 3 5
T4	2 5
T5	1 3 5



Table: C2 |

Itemset	Support
{1,2}	1
{1,3}	3
{1,5}	2
{2,3}	2
{2,5}	3
{3,5}	3



Table: F2 |

Itemset	Support
{1,3}	3
{1,5}	2
{2,3}	2
{2,5}	3
{3,5}	3

Item sets with support less than 2 it are eliminated

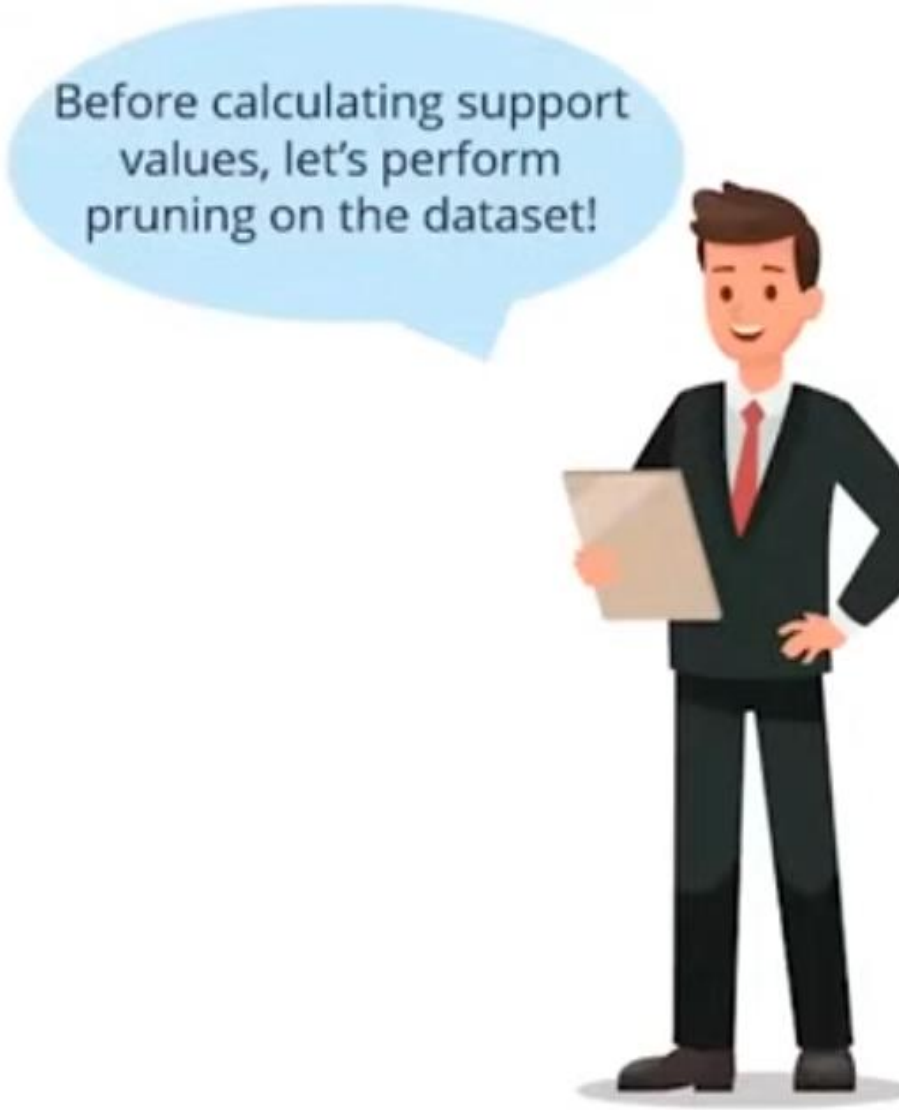
Apriori Algorithm third Iteration

Step 3: Create item sets of size three & calculate their support values.
All the combinations of item sets in F2| are used in this iteration

TID	Items
T1	1 3 4
T2	2 3 5
T3	1 2 3 5
T4	2 5
T5	1 3 5

Table: C3|

Itemset	Support
{1,2,3}	
{1,2,5}	
{1,3,5}	
{2,3,5}	



Apriori Algorithm fourth Iteration

Using the item sets of C3|, create new itemset C4|.

TID	Items
T1	1 3 4
T2	2 3 5
T3	1 2 3 5
T4	2 5
T5	1 3 5



Table: F3|

Itemset	Support
{1,3,5}	2
{2,3,5}	2



Table: C4|

Itemset	Support
{1,2,3,5}	1

Since support of C4| is less than 2, stop and return to the previous itemset, i.e. CI3